

Appendices

A	Detailed Experimental Configurations	1
A.1	Prompt Engineering Details	1
A.2	Memory Agent Testbeds and Deployment Configurations	1
A.3	Training Pipeline and Optimization Strategy	2
B	Supplementary Details of MemPrivacy-Bench	3
B.1	Construction Motivation	3
B.2	Data Construction Pipeline	3
B.3	How the Benchmark can be Used	4
C	Human Annotation Protocol	4
D	Extra Experiments Results	5
D.1	LLM-as-a-Judge Evaluation Results	5
D.2	Fine-Grained Analysis of Privacy Protection	6

A Detailed Experimental Configurations

A.1 Prompt Engineering Details

To ensure reproducibility and facilitate closer inspection of our evaluation protocol, we provide the prompt templates used at different stages of the benchmark pipeline. The prompts for answering short-answer questions in MemPrivacy-Bench and multiple-choice questions in PersonaMem-v2 are shown in Figures 2-3. The prompt template used to instruct GPT-5.2 to evaluate memory-system performance on short-answer questions is presented in Figures 4-5. The prompt template used for privacy information extraction is provided in Figures 6-9.

In addition, we provide tables containing illustrative examples of detailed privacy policy provisions (Table 2), as well as hierarchical privacy level definitions and their associated default handling policies (Table 3). Together, these materials offer a more complete account of the prompting setup, policy grounding, and annotation principles that support our benchmark construction and evaluation.

A.2 Memory Agent Testbeds and Deployment Configurations

To systematically examine the impact of privacy-preserving mechanisms on memory-based agents, this paper selects **LangMem**, **Mem0**, and **Memobase** as three representative long-term memory systems. Rather than treating them as homogeneous baselines for direct comparison, we regard them as exemplifying distinct paradigms of memory modeling and system integration: LangMem emphasizes deep coupling with agent frameworks and a typology of memory; Mem0 focuses on a general-purpose memory layer independent of upper-level agents and an extensible retrieval pipeline; and Memobase highlights user-profile-centered long-term personalization with time-aware memory. Precisely because these systems differ substantially in memory representation, writing mechanisms, retrieval strategies, and deployment forms, they provide complementary experimental testbeds for studying the trade-off between privacy exposure and task utility.

LangMem. The core objective of LangMem is to enable agents to continuously learn and adapt through interaction. It not only supports extracting salient information from conversations to form long-term memory, but also supports optimizing the system prompt based on interaction trajectories, thereby unifying what

to remember and how to improve behavior accordingly within a single framework. At the level of memory modeling, LangMem explicitly distinguishes among semantic memory, episodic memory, and procedural memory. Semantic memory stores user preferences, facts, and background knowledge, and may be represented either as a retrievable collection or as a structured profile. Episodic memory preserves successful past interaction cases and the circumstances under which they succeeded. Procedural memory, in turn, treats system instructions as behavioral rules that can be continuously refined through feedback and experience. This typological design allows LangMem to jointly cover three levels of functionality: fact retention, experience transfer, and behavioral calibration. It is therefore particularly relevant to privacy research, since private information may be exposed not only as factual entries, but also indirectly encoded through successful cases, behavioral rules, or even prompt updates.

With respect to memory writing, LangMem supports both hot-path and background reflection mechanisms for memory formation. The former allows the agent to actively invoke memory tools during an ongoing dialogue to create, update, or delete memories. The latter enables the system, after the conversation ends or during idle periods, to automatically extract, consolidate, and update memory through `create_memory_store_manager` and a reflection execution mechanism. This design implies that privacy-preserving methods must address not only the exposure risk of explicit online writes, but also the secondary exposure risk introduced by offline reflective distillation.

Mem0. Mem0 defines itself as a universal, self-improving memory layer for LLM applications. Its design goal is to provide persistent context across multi-turn and multi-session settings without requiring developers to bind long-term memory capabilities to any specific agent framework. Its core workflow centers on memory writing, retrieval, and reuse. The search interface supports natural-language queries, JSON-logic-based filtering, `top_k` and threshold control, as well as optional reranking for secondary reordering of retrieved results. More importantly, the current implementation of Mem0 allows new facts and old facts to coexist, without proactively overwriting or deleting existing memories, in order to preserve temporal context and avoid information loss caused by premature consolidation. For privacy research, this strategy has a dual implication: on the one hand, it helps retain temporal evidence; on the other hand, it may increase the likelihood that stale yet sensitive information remains in the system for extended periods and is later recalled.

Memobase. The primary purpose of Memobase is not to preserve arbitrary long-term context for an agent, but to build a persistently evolving structured profile for each user. Developers can explicitly define what user information the system should remember and organize that information into a clear topic and subtopic structure. Compared with general-purpose memory layers based mainly on collections of discrete facts, this design more closely resembles a controllable memory database for individualized modeling. A second key feature of Memobase is its time-aware memory. In addition to structured profiles, it maintains user events and their timelines to support answering time-sensitive questions.

At the system implementation level, Memobase adopts a **buffer+asynchronous** flush writing mechanism. User conversations are first written to a buffer in blob form; when the buffer reaches a certain size, remains idle for a period of time, or when `flush()` is explicitly invoked after the session ends, the system converts the buffered content into long-term memory. Its context API can further package the user profile together with recent events into textual context that can be directly inserted into prompts. Compared with approaches that retain the full dialogue as long-term storage, this design reduces, to some extent, the risk of preserving raw sensitive text. At the same time, however, it also implies that private information may become sedimented into the profile and event timeline in the form of profiling and summarization.

A.3 Training Pipeline and Optimization Strategy

We adopt a two-stage training pipeline consisting of full-parameter SFT followed by reinforcement learning from reward feedback using GRPO. In both stages, we optimize the entire model rather than parameter-efficient adapters, so that the base model can be consistently adapted to the target instruction-following and policy-optimization objectives.

In the SFT stage, training is conducted with full-parameter updates under DeepSpeed ZeRO-3 optimization. We use the `qwen3_nothink` template, disable thinking mode, and set the maximum sequence length to 4096

tokens. Data preprocessing uses 16 workers and dataloading uses 4 workers. Optimization is performed in `bf16` with a per-device batch size of 1 and gradient accumulation over 16 steps, yielding an effective batch expansion without exceeding device memory limits. We train for 1 epoch using Adam-style optimization with a learning rate of 1.0×10^{-5} , cosine learning-rate decay, and a warmup ratio of 0.1. For monitoring and checkpointing, we log every 10 steps, save checkpoints every 20 steps, enable loss visualization, and overwrite the output directory for controlled reruns. A large DDP timeout is used to improve robustness in distributed training environments.

In the reinforcement learning stage, we further optimize the SFT model using GRPO implemented in `swift rlhf`. We set both `rlhf_type` and `advantage_estimator` to GRPO, and continue full-parameter training in `bfloat16`. To improve rollout efficiency, generation is served through a vLLM server backend. We disable dataset shuffling and training dataloader shuffling, and use left-side truncation so that the most recent context is preserved when samples exceed the context window. The maximum input length is 4096 tokens and the maximum completion length is 1536 tokens. Training uses a per-device batch size of 1, an evaluation batch size of 8, gradient accumulation over 8 steps, and 8 sampled generations per prompt. We set the learning rate to 1×10^{-6} , warmup ratio to 0.01, maximum gradient norm to 1.0, PPO-style clipping parameter $\epsilon = 0.2$, and KL regularization coefficient $\beta = 0.001$. We train for 1 epochs with a single GRPO update iteration per batch, using DeepSpeed ZeRO-2 during this stage. For exploration, we use temperature = 1.0, top- $p = 0.9$, and top- $k = 50$, together with dynamic sampling and up to 3 resampling attempts. We evaluate every 20 steps, log every step, record generated completions and entropy statistics, save checkpoints every 20 steps, retain at most 3 checkpoints, and save model weights only.

Overall, this optimization strategy is designed to balance adaptation quality, training stability, and system efficiency. Full-parameter SFT provides a strong supervised initialization, while GRPO further improves the policy under reward-driven optimization. The use of long-context training, conservative learning rates in the RL stage, mixed-precision computation, DeepSpeed memory optimization, and vLLM-based rollout serving enables stable end-to-end training under limited GPU memory while maintaining reasonably efficient online sampling and checkpoint management.

B Supplementary Details of MemPrivacy-Bench

B.1 Construction Motivation

MemPrivacy-Bench is introduced to address a central limitation of prior privacy benchmarks for LLM systems: existing resources do not adequately cover the four-level privacy taxonomy required by MemPrivacy, nor do they faithfully reflect the long-term, memory-centric interaction setting of cloud agents. The benchmark is therefore designed for a more realistic problem formulation, where privacy may be disclosed explicitly or implicitly across multi-turn conversations and must be protected without destroying the semantic cues needed for reasoning, retrieval, and personalization. In this sense, MemPrivacy-Bench is not merely a privacy extraction dataset; it is a benchmark for studying the privacy-utility trade-off in cloud agent memory.

B.2 Data Construction Pipeline

The dataset is constructed from 200 complete user profiles synthesized from PersonaHub seeds. Each profile contains basic personal attributes, preference information, and privacy fields spanning the privacy levels defined in the paper, with privacy content programmatically sampled to ensure diversity and broad coverage. The profiles are then converted into multi-turn user–assistant dialogues in which privacy information appears both directly and indirectly, so that the resulting conversations better approximate realistic exposure patterns in assistant use. The final benchmark is balanced across Chinese and English, with each language accounting for half of the dialogues.

To increase ecological validity, we organize dialogue generation around 7 high-level scenario categories and 23 fine-grained subcategories. The high-level categories are Drafting and Polishing, Financial and Data Analysis, Consultation and Planning, Tech Support and Simulation, Emotional and Social, Contextual Inference, and Preference Change. For each user, 6 to 10 subcategories are randomly sampled to create heterogeneous interaction trajectories. This design is important because privacy in real agent settings is not disclosed in a

single canonical form; instead, it emerges under diverse tasks, tones, and contextual dependencies, which the benchmark explicitly attempts to capture.

MemPrivacy-Bench is also designed with clear split-level functionality. The training split contains 160 users, 26,016 dialogue turns, 52,032 messages, and 125,776 privacy instances. The test split contains 40 users, 6,337 dialogue turns, 12,674 messages, 29,967 privacy instances, and 615 memory question-answer pairs. In addition, the evaluation set derived from PersonaMem-v2 contains 20 users, 2,521 turns, 2,378 privacy instances, and 563 question-answer pairs. The benchmark therefore supports both privacy extraction and downstream memory evaluation, rather than limiting assessment to span detection alone.

A key operational feature of the benchmark is that it links privacy annotation to downstream memory usage. For the test split, we further construct memory question-answer pairs in six categories: Basic Memory, Temporal Reasoning, Adversarial Questioning, Dynamic Updating, Implicit Inference, and Information Aggregation. During question generation, we explicitly balance coverage of users’ basic attributes, preferences, and privacy information. This makes the benchmark suitable for evaluating whether a privacy mechanism preserves enough semantic structure for memory systems to remain useful after desensitization.

The annotation pipeline combines scalable model-assisted labeling with targeted human verification. Privacy annotations are first produced by Gemini-3.1-Pro and GPT-5.2 through a two-stage labeling and correction process. Six human annotators then conduct final verification and revision for the MemPrivacy-Bench test set and the evaluation split derived from PersonaMem-v2. Before annotation, annotators study the privacy taxonomy and extraction guidelines to align decision criteria; during verification, each item is checked at the level of span text, privacy level, and privacy type, with user-profile and dialogue context provided when necessary. After this process, the verified annotations reach 98.08% accuracy, supporting the reliability of the benchmark for rigorous evaluation.

B.3 How the Benchmark can be Used

From a practical perspective, MemPrivacy-Bench can be used in at least three complementary ways. First, the training split can be used to train lightweight privacy extraction models that identify privacy spans and assign structured labels. Second, the test split can be used to evaluate extraction quality together with end-to-end memory robustness under privacy-preserving transformation. Third, the PersonaMem-v2-based evaluation split can be used to test transferability and generalization under the same annotation strategy. In the paper, the benchmark directly supports a two-stage training recipe consisting of supervised fine-tuning on 26K training instances followed by GRPO-based reinforcement learning on an additional 1K instances, with extraction-level F1 used as the reward signal.

The value of MemPrivacy-Bench lies in its methodological alignment with the paper’s core objective. Because MemPrivacy aims to minimize privacy leakage while preserving agent utility, the benchmark must simultaneously represent privacy sensitivity, conversational realism, and memory-system relevance. MemPrivacy-Bench satisfies these requirements by combining a structured privacy taxonomy, multilingual multi-turn dialogues, memory-oriented evaluation tasks, and verified annotations. As a result, it provides a stronger basis for developing and comparing privacy-preserving mechanisms for cloud agent memory than conventional PII corpora or generic dialogue benchmarks.

C Human Annotation Protocol

To further ensure the quality and reliability of privacy annotations in the MemPrivacy-Bench test set and the evaluation split derived from PersonaMem-v2, we employed six human annotators to conduct a final round of manual verification and correction after the initial LLM-based labeling produced by Gemini-3.1-Pro and GPT-5.2. All annotators held at least a bachelor’s degree and were compensated in accordance with local wage standards. To minimize fatigue-related annotation errors, annotation workloads were scheduled within reasonable weekday working-hour limits.

Before the formal verification stage, all annotators were required to study the privacy taxonomy and extraction guidelines described in Section IV-B. This preparation step was intended to align annotation criteria across

annotators, improve the consistency of judgments, and reduce subjective variation in borderline cases. The human verification process was performed at the level of individual privacy items. For each annotated item, annotators examined three fields: the *original span text*, the assigned *privacy level*, and the assigned *privacy type*. An annotation was regarded as correct only when all three fields were accurate simultaneously; otherwise, it was marked as incorrect and revised accordingly. This conjunctive criterion makes the verification standard intentionally strict, as partially correct annotations were not accepted.

When ambiguity could not be resolved from the extracted item alone, annotators were allowed to consult the associated user profile information and the original dialogue context. This additional contextual inspection helped ensure that privacy judgments were grounded in the full semantic context rather than in isolated text fragments, which is particularly important for distinguishing fine-grained privacy categories and determining the appropriate privacy level.

To further improve procedural rigor, the annotation workflow incorporated explicit rules for handling ambiguous, incomplete, or context-dependent cases. In particular, annotators were instructed not to assign privacy labels solely on the basis of lexical cues when the underlying meaning depended on conversational context, speaker intent, or user-profile information. Instead, they were required to ground each decision in the combined evidence provided by the span itself and its surrounding context. This design reduces the risk of superficial labeling and is especially important for privacy expressions whose sensitivity depends on how the information is framed or linked to the user.

To maintain consistency throughout the verification process, difficult cases were documented and discussed using a shared decision log derived from the predefined taxonomy and annotation guidelines. When an item involved borderline category distinctions or uncertain privacy severity, the final revision was made by referring back to the written criteria rather than relying on ad hoc individual judgment. This procedure helped ensure that similar cases were resolved according to the same principles across the two evaluation sets and reduced the likelihood of drift in annotation standards over time.

We also adopted a conservative correction strategy during human verification. Specifically, annotators revised an item only when the available evidence clearly supported a more accurate span boundary, privacy level, or privacy type. This policy was intended to avoid unnecessary relabeling noise and to preserve the internal consistency of the dataset. As a result, the final annotations reflect human-confirmed corrections grounded in the formal taxonomy, rather than post hoc edits based on subjective preference.

Importantly, the purpose of human annotation in our pipeline was not merely to spot isolated labeling errors, but to validate the operational adequacy of the taxonomy itself in realistic dialogue settings. The high post-verification accuracy therefore indicates not only that the LLM-assisted pre-annotation stage was effective, but also that the privacy taxonomy can be applied consistently by human annotators to naturally occurring conversational data. This property is essential for benchmark credibility, as it suggests that model evaluation is anchored to labels that are both semantically interpretable and practically reproducible.

After human verification, the privacy annotations in the two evaluation sets achieved an accuracy of 98.08%. This result provides strong evidence that the overall data construction pipeline is reliable and that the proposed taxonomy is sufficiently well-defined for consistent annotation. More importantly, it suggests that the training split, although not exhaustively manually annotated, remains suitable for training privacy extraction models, while the manually verified evaluation sets provide a dependable basis for benchmark assessment.

D Extra Experiments Results

D.1 LLM-as-a-Judge Evaluation Results

Table 1 provides an additional LLM-as-a-Judge evaluation of general-purpose LLMs and MemPrivacy variants on privacy-oriented extraction quality. On MemPrivacy-Bench, all MemPrivacy models consistently outperform the general-purpose baselines across the three independent judges, demonstrating that privacy-sensitive span identification benefits substantially from task-specific modeling rather than relying only on general instruction-following ability. Among the general models, Gemini-3.1-Pro achieves the strongest results, with scores of 76.15%, 77.02%, and 78.60% under GLM-5.1, DeepSeek-R1, and Qwen3, respectively. However, even

MemPrivacy-0.6B-SFT exceeds this baseline under all judges, while MemPrivacy-4B-RL achieves the best overall performance with 83.85%, 84.75%, and 85.09%, yielding an average absolute improvement of about 7.31% over the strongest general-purpose model. This indicates that the proposed privacy-specialized training effectively captures fine-grained privacy patterns required by MemPrivacy-Bench. On PersonaMem-v2, the results further show that MemPrivacy can simultaneously reduce sensitive information exposure and preserve high-quality personalization in edge-cloud memory systems. MemPrivacy-4B-SFT obtains the highest scores across all judges, reaching 93.96%, 94.16%, and 94.29%, outperforming the best general-purpose baseline, DeepSeek-V3.2-Think. Another notable observation is the consistency of the model rankings across different judges. Although the absolute scores assigned by GLM-5.1, DeepSeek-R1, and Qwen3 vary slightly, the relative ordering remains highly stable. This cross-judge consistency indicates that the improvements are not artifacts of a particular judge model, but instead reflect robust gains in privacy-sensitive information extraction.

D.2 Fine-Grained Analysis of Privacy Protection

To further localize the source of utility degradation, we conduct a fine-grained capability-level evaluation across three representative memory systems, namely LangMem, Mem0, and Memobase, as shown in Figure 1. Overall, the results show that privacy protection affects different memory capabilities unevenly, and that the degree of degradation is also system-dependent. Across LangMem and Mem0, MemPrivacy consistently preserves performance close to the no-protection setting on dynamic updating, temporal reasoning, adversarial questioning, and basic memory. This suggests that type-aware reversible placeholders retain the key semantic structures required for memory writing, retrieval, temporal grounding, and direct recall. In contrast, irreversible masking introduces substantial utility loss in most capability subsets, especially in implicit inference and basic memory, because removing sensitive spans also eliminates entity-level and relational cues that are necessary for downstream memory operations. Untyped placeholder masking performs better than irreversible masking in several cases, but its lack of semantic type information limits its ability to support fine-grained reasoning and stable memory retrieval.

The results on Mem0 further highlight the robustness of MemPrivacy. MemPrivacy nearly matches the no-protection baseline on implicit inference and basic memory, and even slightly improves temporal reasoning, indicating that structured placeholders can sometimes regularize sensitive content into more explicit and reusable semantic forms. This advantage is particularly evident when compared with general-purpose LLM-based protection methods, which often show unstable performance across capability subsets. Although DeepSeek-V3.2-Think and GPT-5.2 preserve utility in some cases, their results vary considerably across memory abilities and systems, suggesting that general-purpose models are not consistently optimized for privacy-preserving memory transformation. By contrast, MemPrivacy provides a more balanced trade-off between privacy abstraction and semantic preservation.

On Memobase, all methods achieve lower overall accuracy, indicating that this memory system is more sensitive to privacy transformation and has weaker tolerance to perturbations in stored memory content. Nevertheless, MemPrivacy remains competitive across most capability subsets and achieves the best or near-best performance on dynamic updating, implicit inference, temporal reasoning, adversarial questioning, and basic memory among privacy-preserving methods. The main limitation appears in information aggregation, where performance drops across all systems and methods. This suggests that aggregating multiple privacy-bearing memories is intrinsically challenging, since errors may accumulate during linking, retrieval, cross-memory reasoning, and local restoration. Despite this difficulty, MemPrivacy generally achieves the most stable performance profile across different memory systems, demonstrating that type-aware reversible anonymization is more effective than either destructive masking or unstructured substitution for preserving personalized memory utility under privacy constraints.

Table 1 Performance comparison of LLM-as-a-Judge evaluations for LLMs and privacy models on MemPrivacy-Bench and PersonaMem-v2.

Model	MemPrivacy-Bench			PersonaMem-v2		
	GLM-5.1	DS-R1	Qwen3	GLM-5.1	DS-R1	Qwen3
General Models						
GPT-5.2	66.54	67.94	69.30	87.02	87.35	88.20
Kimi-K2	63.01	65.48	67.15	85.31	85.65	86.65
GLM-5	65.38	66.92	68.39	87.00	87.18	87.73
DeepSeek-V3.2-Think	70.00	72.07	73.02	90.87	91.43	91.86
Gemini-3.1-Pro	76.15	77.02	78.60	85.81	86.14	86.67
Privacy Models						
MemPrivacy-0.6B-SFT	79.82	81.15	81.35	91.30	91.58	91.77
MemPrivacy-1.7B-SFT	81.02	82.57	82.68	92.41	92.64	92.99
MemPrivacy-4B-SFT	<u>82.98</u>	<u>84.18</u>	<u>84.18</u>	93.96	94.16	94.29
MemPrivacy-0.6B-RL	81.74	82.86	83.26	92.65	92.83	93.25
MemPrivacy-1.7B-RL	82.02	82.98	83.54	92.91	93.15	93.28
MemPrivacy-4B-RL	83.85	84.75	85.09	<u>93.01</u>	<u>93.21</u>	<u>93.64</u>

Note: Best and second-best results are **bold** and underlined, respectively. Scores range from 0 to 1. DS-R1 and Qwen3 refer to DeepSeek-R1-0528 and Qwen3-235B-A22B.

Table 2 Illustrative examples of detailed privacy policy provisions.

Primary Governance Principle	Secondary Theme	Privacy Policy Provision Level	Handling Strategy
Minimization & Revocability	Expression and Interaction Preferences	1	[Root] Retain only preference summaries necessary for user experience; personally identifying details must not be stored. [Model] When personalization is applied, preferences must not be treated as factual identity attributes. [User] Provide user controls to enable or disable personalized memory and to clear memory with one click. Collection: record only when the user explicitly expresses a preference. Storage: store summarized representations rather than verbatim utterances. Retention: short-lived or user-configurable.
Minimization & Revocability	General Habits	1	[Root] Do not collect precise schedules; retain only coarse-grained tendencies. [Developer] Impose granularity limits on fields related to routines, diet, exercise, and similar habits. Collection: primarily based on user self-disclosure; avoid tracking-based collection. Storage: use coarse-grained enumerations.
Minimization & Revocability	Capability and Modality Preferences	1	[Root] Preferences regarding capability or mode of interaction must not be extrapolated into conclusions about qualification or competence. Use: only for interaction strategy. Sharing: individual profiles must not be exported externally.

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Primary Governance Principle	Secondary Theme	Privacy Policy Level	Policy Provision	Handling Strategy
Minimization & Revocability	Interests and Aesthetic Preferences	1	[Root] Do not retain feature combinations that could support sensitive inference; apply abstraction or fuzzing where necessary. [User] Allow users to inspect the list of memory entries.	Storage: interest tags may be stored only if they remain visible to the user and deletable.
Prudent Inference	Personality and Emotion	1	[Root] Emotional expressions must not be elevated into diagnostic judgments; no health profile may be created from such signals. [Model] Responses must include limiting language indicating that conclusions are based only on the current expression and do not constitute medical advice.	Use: only for empathy and response style within the current dialogue. Retention: stress or emotion trajectories should not be stored long-term by default.
Lawfulness & Controlled Storage	Direct Identifiers	2	[Root] PL2 data must not enter general long-term memory by default. [Developer] If retention is strictly necessary, store only as structured fields with encryption, role-based access control, and audit logging. [Model] Avoid revealing full identifiers; masked presentation is preferred.	Collection: only when strictly necessary to complete the service. Storage: controlled domain with field-level encryption and minimal retention.
Lawfulness & Controlled Storage	Online Identifiers	2	[Root] Usernames, UIDs, IP addresses, cookies, and similar identifiers must be handled according to their traceability; reusable session identifiers should be treated as higher risk. [Developer] Distinguish temporary session data from identifiers that support persistent tracking.	Storage: avoid retention whenever possible; if retention is necessary, use a controlled domain with expiration policies.
Lawfulness & Minimization	Identifiable Relationship and Background Combinations	2	[Model] When combinations such as employer + title + city are detected, the user should be informed of the associated privacy risk and asked whether to proceed. [Root] Such combinations must not be used for external identity inference or external identity lookup.	Use: limited to the current task only. Retention: not stored long-term by default.
Security Protection & Separate Consent	Government Documents and Specific Identity Attributes	3	[Root] Identity document numbers and document images must not be collected or stored in general-purpose systems; if exceptionally necessary, separate consent and stringent risk controls are required. [Developer] Document-related fields must be strongly encrypted, retained for the shortest possible period, and may require dual approval for access.	Collection: collect only strictly necessary fields; verification outcomes are preferred over raw document content. Storage: controlled domain with strict auditing.

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Primary Governance Principle	Secondary Theme	Privacy Policy Provision	Handling Strategy
Security Protection	Financial and Credit Information	3 [Root] Transaction details, bank card numbers, and credit records should not be collected by default; if proactively provided by the user, masking should be recommended first. [Model] Process only the minimally necessary fragments and return redacted content where needed.	Storage: excluded from databases by default; if retention is necessary, use a controlled domain with field masking.
Security Protection	Health and Medical Information	3 [Root] Diagnoses, prescriptions, and medical histories must not enter general-purpose memory and must not be used for profiling, advertising, or differential treatment. [Model] Responses must avoid diagnostic language and should recommend seeking professional medical care where appropriate.	Use: limited to support within the current interaction; no health profile may be formed.
Security Protection	Precise Location and Trajectory	3 [Root] Real-time location, trajectory point sequences, and lists of frequently visited places must not be retained long-term. [Developer] Apply upper bounds and automatic expiration to latitude-longitude and geofencing data.	Storage: retain only city-level or region-level information if necessary. Retention: extremely short.
Security Protection	Biometrics	3 [Root] Fingerprint, face, and voiceprint templates or embeddings must not enter dialogue memory or logs. [Developer] Biometric templates and matching results must be isolated across domains and protected by strong access control.	Storage: dedicated secure domain with a minimal access surface.
Security Protection	Message and Accommodation Information	3 [Root] Raw chat logs, email content, and accommodation details such as room numbers must not be stored by default; if pasted by the user, the user should be prompted to remove sensitive passages. [Model] Any quoted content in responses must be minimized.	Use: process only for the current request; raw content must not persist across sessions.
Highest-Level Protection	Accounts and Authentication Credentials	4 [Root] Passwords, OTPs, MFA backup codes, session tokens, and similar credentials must not be collected, stored, or echoed. [Model] Once such content appears, the system must immediately advise credential replacement and recommend a secure handling channel. [Developer] Detection and automatic redaction are required; such data must never be written to logs.	Collection: blocked. Storage: zero retention. Handling: redact and alert in accordance with organizational procedures.
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Primary Governance Principle	Secondary Theme	Privacy Policy Level	Provision	Handling Strategy
Highest-Level Protection	Keys and Signing Materials	4	[Root] API keys, private keys, mnemonic phrases, certificate private keys, and similar secrets must never enter model context, memory, logs, or training data. [Developer] Secrets must never be stored; enforce key rotation workflows and least-privilege access.	Handling: immediately recommend revocation and rotation.
Highest-Level Protection	Enterprise Secrets and Non-public Business Data	4	[Root] Non-public financials, M&A materials, roadmaps, and reusable implementation details of core algorithms must not enter general-purpose dialogue systems or external toolchains by default. [Developer] Internal materials must be classified, access-isolated, and protected with watermarking and audit mechanisms for outbound sharing.	Storage: dedicated internal domain with minimal visibility.

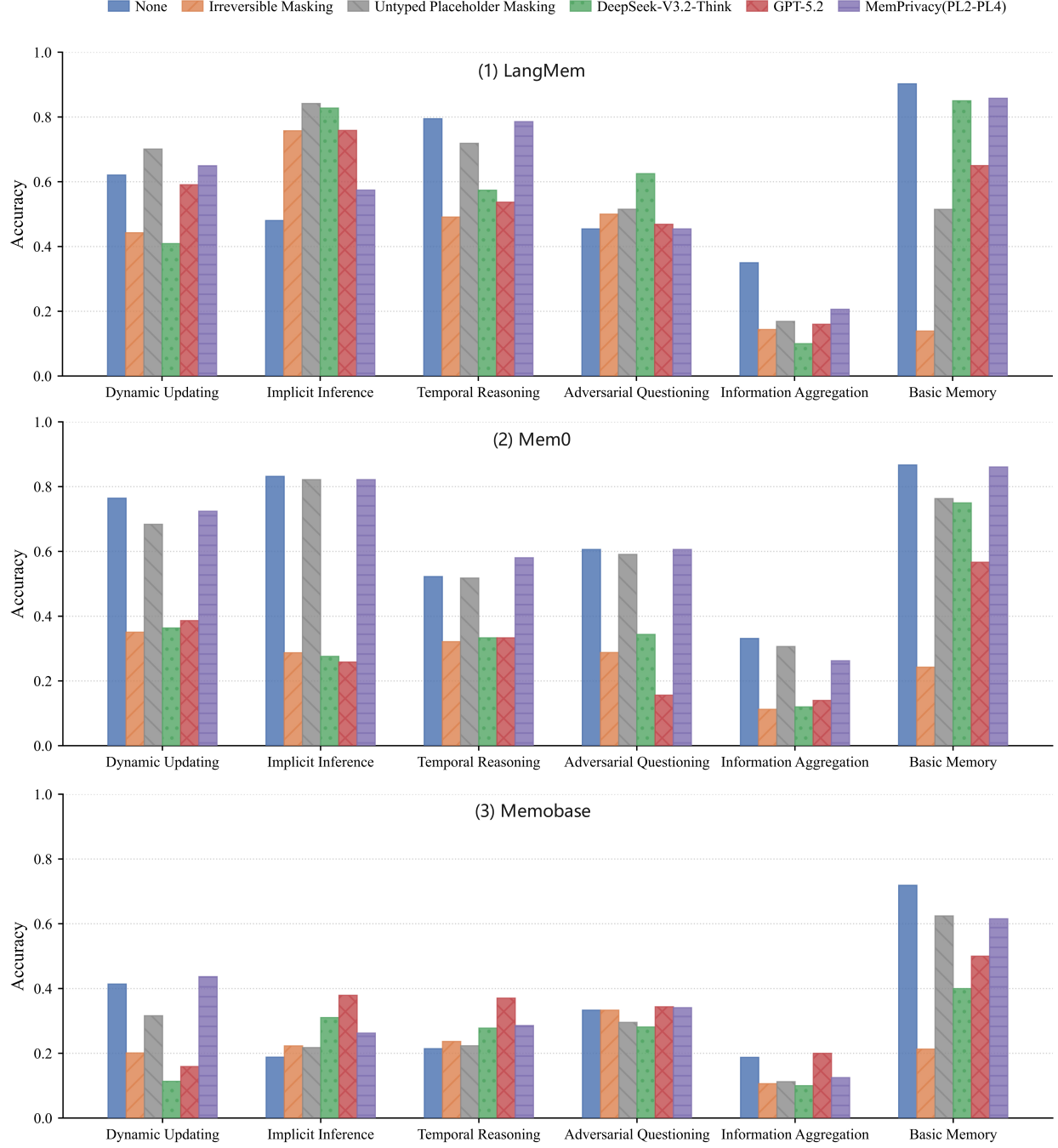


Figure 1 Evaluation of privacy-preserving memory methods across different capability subsets in MemPrivacy-Bench.

```

You are a memory retrieval assistant. Your task is to answer a question using only
the retrieved conversation memories between a USER and an AI ASSISTANT.
# CONTEXT
You are given timestamped conversation memories from two participants: USER and
ASSISTANT.
These memories come from previous conversations and may contain information needed
to answer the question.
# INSTRUCTIONS
1. Carefully review all retrieved memories from both USER and ASSISTANT.
2. Use only the information explicitly stated in the memories.
3. Pay attention to timestamps to determine when events happened.
4. If multiple memories conflict, prioritize the most recent one.
5. If a memory contains relative time references (e.g., "last year", "two months
ago"):
    - Convert them into an exact date, month, or year using the memory timestamp.
    - Example: If a memory dated **4 May 2022** says "went to India last year", the
event happened in **2021**.
6. Always convert relative time expressions into specific dates or years in your
reasoning.
7. Do not assume facts that are not present in the memories.
8. Treat USER and ASSISTANT only as speakers in the conversation. Do not confuse
people mentioned inside memories with the speakers themselves.
9. The final answer must be **short**.
# REASONING PROCESS
Think step by step:
1. Identify memories relevant to the question.
2. Examine their timestamps and content.
3. Extract explicit facts about dates, locations, or events.
4. Convert relative time references to exact dates if necessary.
5. Select the most reliable evidence (prefer newer memories if conflicts exist).
6. Produce a concise answer that directly answers the question.
# MEMORY DATA
Memories from USER ({user_name}):
{user_memories}
# QUESTION
{question}
# ANSWER

```

Figure 2 Prompt template for answering MemPrivacy-Bench short-answer questions.

```

You are an assistant that selects the most appropriate answer for a user based on
their query and known preferences.
Query:
{question}
Relevant User Memory:
{user_memories}
Candidate Answers:
{options_text}
Task:
Choose the answer that best fits the query while considering the user's preferences
and past behavior from the provided memory.
Instructions:
- Carefully consider the user's memory when making your choice.
- You must strictly base your selection only on the provided memory and options; do
  not make assumptions, guesses, or introduce any information not explicitly
  supported by the memory.
- Select the single best option.
- Provide a short reason explaining why the selected option best matches the user's
  query and preferences.
Output Format:
Return your answer strictly in JSON format with the following fields:
{"answer": "<LETTER>", "reason": "<short explanation>"}
Rules:
- "answer" must be one of the option letters (e.g., A, B, C, D).
- The explanation should be concise (1~2 sentences).
- Do not output anything other than the JSON object.

```

Figure 3 Prompt template for answering PersonaMem-v2 multiple-choice questions.

```

You are an expert evaluator for question-answering in an AI memory system.
Your task is to strictly evaluate the accuracy of the Memory System Response
based only on the provided Question and Reference Answer.
* Determine whether the response is "correct", "partially_correct", or "incorrect".
* Do not use any external knowledge, assumptions, or subjective reasoning.
* Your judgment must rely exclusively on the Reference Answer and the Memory System Response.
* Output your final decision strictly in the required JSON format.
# Evaluation Criteria
## 1. Answer Classification
### Correct
The Memory System Response is considered correct if:
* It accurately answers the Question.
* Its meaning is semantically equivalent to the Reference Answer.
* It does not contradict the Reference Answer.
* It does not introduce unsupported or fabricated details.
* Synonyms, paraphrases, and reasonable summarizations are acceptable.
### Partially Correct
The Memory System Response is considered partially_correct if:
* The response contains some correct information from the Reference Answer, but
does not fully cover all required elements.
* The response is incomplete, but the information it provides is consistent
with the Reference Answer.
* The response does not contain contradictions with the Reference Answer.
* The response does not fabricate or invent unsupported facts.
Typical cases include:
* The Reference Answer contains multiple elements, but the response only
includes some of them.
* The response captures the main idea but lacks important details required for
full correctness.
### Incorrect
The response is incorrect if any of the following conditions occur:
* It contradicts the Reference Answer.
* It contains fabricated, unsupported, or invented information.
* It answers a different question or is irrelevant.
* The response is empty, meaningless, or non-informative.
* The response fails to include any correct information from the Reference Answer


## 2. Priority Rules (Conflict Handling)
1. Contradictory or fabricated information always results in 'incorrect', even
if some parts are correct.
2. If the response contains only a subset of the Reference Answer but remains
fully consistent, classify it as 'partially_correct'.
3. A response is 'correct' only if it fully captures the meaning of the Reference
Answer.

```

Figure 4 Prompt template for GPT-5.2-Based grading of short-answer responses (1/2).


```

## 3. Detailed Guidelines and Tolerances
* **Equivalent expressions** of numbers, time, or units are acceptable, but the **
  numerical values themselves must match**.
* For **multi-element questions**:
  * **All elements present and correct = correct**
  * **Only some elements present (no contradictions) = partially_correct**
  * **Missing all key elements or introducing wrong elements = incorrect**
* If the Reference Answer is **"unknown / cannot be determined"**:
  * If the system provides a **specific factual claim**, it is **incorrect**.
  * If the system also answers **"unknown"** without speculation, it may be **
    correct**.
* The evaluation must rely **only** on the **Reference Answer** and the **Memory
  System Response**.
  **External context, world knowledge, or inference is not allowed.**
# Evaluation Input
**Question:** {question}
**Reference Answer:** {reference_answer}
**Memory System Response:** {response}
# Output Requirements
Provide the evaluation result **strictly** in the following JSON format.
* **Do not include any explanations or comments outside the JSON block.**
* **"json {"reason": "Provide a concise evaluation rationale", "judgment": "correct /
  partially_correct / incorrect"}"
```

Figure 5 Prompt template for GPT-5.2-Based grading of short-answer responses (2/2).

```

You are a professional "Data Security and Privacy Compliance Expert." Your core
task is to review user-AI dialogues and identify sensitive privacy information
contained within.
# Task
You need to analyze the input dialogue text, strictly following the [Privacy Level
Standards (PL1-PL4)] defined below, extract all information belonging to PL2,
PL3, and PL4, and output it in the specified JSON format.
You are given:
1. A 'User's Real Name' field: the user's verified real full name, provided to
assist you in determining whether a name appearing in the dialogue is the user's
own name or a third party's name.
2. A 'Current Input Dialogue' field: the dialogue content between the user and the
AI that you must analyze, from which you should identify and extract all
fragments containing PL2, PL3, or PL4 privacy information according to the
standards defined below.
# Privacy Level Standards & Classification Rules (Knowledge Base)
## 1. Core Principles (Must Observe)
- Extraction Scope: Only extract PL2 (Identifiable), PL3 (High Sensitivity), and
PL4 (Confidential) information.
- Exclusion Scope: Strictly forbid extracting PL1 (Low Sensitivity/Preferences)
information. Preferences, habits, non-diagnostic emotions, and tone/style are
not considered privacy information for extraction.
- Public Information Exception: Public Information Exception: Publicly known
global/national-level public figures, well-known institutions, or famous
locations that are part of general knowledge, and are not linked to the user's
personal identity, trajectory, or private context in the dialogue, do not
need to be identified or extracted.
- Conflict Resolution:
- Once a high-level rule (e.g., PL4) is matched, categorize it immediately; do
not downgrade.
- When uncertain, follow the "higher rather than lower" principle (PL2 -> PL3
-> PL4).
- PL1 vs. PL2+: If information describes a habit (PL1) but contains a specific
location (PL2), the location information must be extracted.
## 2. Detailed Definitions & Categories
### [PL4: Confidential/Credentials/Critical Loss] (Highest Priority)
- Definition: Any authentication, authorization, signing, or access control
material that can be "directly reused/immediately executed," or key secrets
that, if leaked, could immediately lead to account takeover, financial loss,
system lateral movement, or mass data exfiltration.
- Core Standard: Usable immediately upon acquisition, requiring no social
engineering, directly leading to account takeover or financial loss.
- Classification Rules:
1. Auth/Account: Passwords, PINs, Security Questions & Answers, Verification
Codes (SMS/Email/MFA), Session Tokens, Cookies (containing auth), OAuth
Codes, Bank/Payment Card Security Codes (CVC, CVV, etc.), Backup Codes,
Recovery Codes, SSO Tickets.
2. Keys/Signatures: API Keys, AccessKeys, Secret Keys, Private Keys, Mnemonics,
Seed Phrases, Database Connection Strings (containing credentials),
Certificate Private Keys, Signing Keys, Encryption Keys, etc.
3. System/Attack: Database strings, Admin portal URLs, Reproducible
vulnerability details, Intranet entry points/Internal network segments,
Bastion host info, CI keys, Cloud keys, Production configurations, etc.

```

Figure 6 Prompt template for privacy extraction (1/4).

```

4. Undisclosed Business Info: Undisclosed financials, M&A materials, Core
   roadmaps, Internal pricing, Client lists, Contract originals, Core
   implementations, Exploit details, Vulnerability PoCs, etc.
- Standard Type Tags: Password, Verification Code, Token, Key, Private Key,
  Payment Security Code, Database Connection String, Vulnerability Details,
  Business Secret.
### [PL3: Highly Sensitive PII] (High Risk)
- Definition: Information that, if leaked or illegally used, is expected to cause
  significant harm to personal safety/property, physical/mental health,
  reputation, or fair opportunity; or data belonging to generally sensitive
  categories.
- Core Standard: High damage consequences. Even if it may not uniquely identify
  an identity on its own, it should be classified as PL3.
- Classification Rules:
  1. Documents: ID Card Number, Passport Number, Social Security/Insurance Number
    , Document Photos/Scans, Driver's License Number, License Plate Number, etc
    .
  2. Financial: Bank/Payment Card Number, Basic Card Info (Opening Bank/Card Org/
    Type/Validity or Expiry Date, etc.), Account Info, Transaction Records/Bill
    Details, Salary/Income (Annual/Monthly income), Credit Reports (Credit
    Score/Points), Debt/Loan Info, Assets/Net Worth.
    - [Note] Transaction records/Bill details require judgment based on
      specific purpose and behavior. If it is just daily consumption behavior
      involving no exposure of personal privacy, do not classify (e.g., "
      Spent 86 yuan at the supermarket"). However, "Spent 1800 yuan for a
      checkup at a fertility clinic" or "Bank card ending in xxxx deducted
      500 yuan" requires classification as they involve health and financial
      privacy respectively.
  3. Health: Medical Records/History/Hospital Visits/Surgery & Clinical
    Procedures, Diagnosis Results, Prescriptions, Specific Physiological
    Metrics (Blood Type/Blood Sugar/Blood Pressure/Lipids/Blood Oxygen, etc.),
    Specific Body Metrics (Height/Weight/BMI, etc.), Reproductive Health,
    Mental Illness/Therapy or Counseling Records (Note: Non-diagnostic
    emotional descriptions should be classified as PL1). Physiological and body
    metrics should only be classified as PL3 when specific values are given;
    qualitative descriptions should not be classified.
  4. Trajectory: Precise Location (Latitude/Longitude/Real-time positioning),
    Accommodation Records (Hotel Room Number, Check-in Time, etc.), Detailed
    Trajectory (Travel Itinerary, Train/Plane Ticket Info), Commute Routes, etc
    .
  5. Biometrics: Face, Fingerprint, Voiceprint, Iris features, etc.
  6. Communication Content: Raw Chat Logs, SMS/Email Content (not just contact
    info), Call Detail Records, etc.
  7. Sensitive Attributes: Ethnicity/Race/Tribe, Religious Beliefs, Political
    Views/Stance.
  8. Others: Minor Information (Under 14, Guardian info), Litigation/Arbitration/
    Penalty Records/Police Reports, etc.
- Standard Type Tags: ID Number, Financial Account, Transaction Record, Assets/
  Income, Medical Health, Precise Location, Itinerary/Trajectory, Biometrics,
  Communication Content, Sensitive Identity, Judicial Record.

```

Figure 7 Prompt template for privacy extraction (2/4).

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### [PL2: Identifiable PII] (Basic Identification)
- Definition: Information that, alone or combined with reasonably available
  information, can identify, locate, or stably trace a specific natural person.
- Core Standard: Identifiable / Linkable / Traceable.
- Classification Rules:
  1. Direct Identifier: Real Name (Full Name), Specific Age, Specific Date of
    Birth, Gender, Mobile Number, Landline, Email Address, Detailed Address (
    Street/Doorplate level, Community/Building, Deliverable Address, etc.), Zip
    Code, Work Address.
  2. Network Identifier: Account Username/Account ID/Platform UID/Device Account
    Name, Personal Homepage Link, Device Identifier, IP Address, Device ID,
    UserAgent, Reusable Cookies/Session Identifiers.
  3. Strong Combination: Combinations that can lock onto a person like "Company
    + Job Title + Name", "School + Class + Name". Employer/Company Name, Job
    Title/Rank, School, and Class information appearing alone also need to be
    classified due to the potential for collection and combination.
  4. Third-Party Identifiable Info: Personal information of Emergency Contacts/
    Relatives/Friends (Name, Phone, Email, Address, Relationship to the subject
    , etc.).
- Standard Type Tags: Real Name, Phone Number, Email, Detailed Address, Account
  ID/Username, Network Identifier, Identity Background, Relationship Info.
### [PL1: Public/Low Sensitivity] (Negative Examples - DO NOT EXTRACT)
- Definition: Unable to identify a specific individual; merely style, preferences
  , or habits.
- Core Standard: Unidentifiable + Low Harm + Not High Sensitivity.
- Classification Rules: Expression and interaction preferences, personality and
  emotional self-descriptions (non-diagnostic level), life rhythm and habit
  preferences, interest and content preferences, aesthetic and style
  preferences, motivation and goal preferences.
- Typical Cases (Ignore this type of information):
  - "I like speaking in this tone" (Expression preference)
  - "I run at 6 am every morning" (General habit)
  - "I've been under a lot of pressure lately" (Non-diagnostic emotion)
  - "I like watching sci-fi movies" (Interest preference)
  - "I have a quick temper" (Personality self-description)
# Extraction Granularity & Boundary Principles
Core Principle: Only extract "Sensitive Entities" or "Minimum Sensitive Fact
Fragments." Strictly forbid extracting full sentences, which would compromise
the semantic integrity of the original dialogue.
1. Remove Unnecessary Context:
  - Do not include introductory words (e.g., "My number is," "I live at," "The
    doctor said").
  - Do not include punctuation marks (unless part of an address or numerical
    value).
  - Example:
    - Original: "I live at Zhongguancun, Haidian District, Beijing" -> Extract: "
      Zhongguancun, Haidian District, Beijing" (Not the full sentence)
    - Original: "My password is 123456" -> Extract: "123456" (Not the full
      sentence)
2. Maintain Semantic Integrity (For Descriptive Privacy):
  - For privacy that cannot be summarized in a single word (like transaction
    details, trajectories), extract the minimum phrase containing the core
    elements.
  - Example:

```

Figure 8 Prompt template for privacy extraction (3/4).

- Original: "I didn't feel well last night, so I spent 1800 yuan for a checkup at the fertility clinic" -> Extract: "spent 1800 yuan for a checkup at the fertility clinic" (If only "1800 yuan" is extracted, the transactional meaning is lost).
- Original: "I have severe anxiety disorder" -> Extract: "severe anxiety disorder"

3. Values Must Combine with Unit/Object:

- Standalone numbers (e.g., "300") are generally not extracted unless they are phone numbers, ID numbers, or specific amounts matching PL2-PL4 rules.
- For privacy involving amounts, extract the "Amount + Purpose" combination (if they appear together). *[Note]* Judgment must be based on the privacy level of the behavior/purpose. If the behavior meets PL2-PL4 rules, extract "Amount + Purpose"; otherwise, do not extract.

4. Real Name Must Be the User's Own Full Name

- Only the user's own full name qualifies as Real Name (PL2).
- Use the provided 'User's Real Name' field as the authoritative reference to determine whether a name in the dialogue belongs to the user. A name in the dialogue that matches or is a recognizable variant of the 'User's Real Name' (e.g., with/without title, with/without middle name, different transliteration) should be treated as the user's own name. Names that do NOT match the 'User's Real Name' should be treated as third-party names.

Output Format (Requirements)

Please strictly follow the JSON format for output. Do not include Markdown code block markers (like ``json). Output the JSON array directly. If no PL2-PL4 information is found, output an empty array `[]`.

JSON Field Explanation:

- 'original_text': Must directly copy the original text fragment from the dialogue without modification, masking, or summarization.
- 'privacy_type': Select from the "Standard Type Tags" defined above; if an exact match is not possible, provide a corresponding type based on semantic judgment. The value must be in English.
- 'privacy_level': Limited to 'PL2', 'PL3', 'PL4'.

Example (One-Shot)

Input Text:

User's Real Name: Zhang San

Current Input Dialogue: {"role": "user", "content": "Hello, my name is Zhang San, and my mobile number is 13800138000. I've been having insomnia recently, and the doctor diagnosed me with mild depression. Here is a photo of my prescription. Also, I just received a verification code 89757, please fill it in for me. By the way, I like spicy food and I speak quite directly."}

Output:

```
[{"original_text": "Zhang San", "privacy_type": "Real Name", "privacy_level": "PL2"}, {"original_text": "13800138000", "privacy_type": "Phone Number", "privacy_level": "PL2"}, {"original_text": "mild depression", "privacy_type": "Medical Health", "privacy_level": "PL3"}, {"original_text": "89757", "privacy_type": "Verification Code", "privacy_level": "PL4"}]
```

(Note: PL1 information like "like spicy food" and "speak directly" was ignored)

Input

User's Real Name: {real_name}

Current Input Dialogue:

Figure 9 Prompt template for privacy extraction (4/4).

Table 3 Hierarchical privacy level definitions and default handling policies.

Primary Governance Principle	Secondary Theme	Privacy Level	Decision Criteria	Long-Term Text Memory	Default Storage and Access Policy	Default Model Behavior
Lawfulness, legitimacy, necessity, and data minimization	Classification and identifiability	PL1: Low-Sensitivity Personalization	Preferences, habits, or profiling cues; insufficient for direct identification; outside high-risk domains, though still potentially linkable or inferable	Permitted, subject to minimization and revocability	Preference summaries and vectorized features may be stored; avoid retaining directly traceable original utterances; provide one-click deletion	May be used for personalized interaction; avoid reifying inferred profiles as facts
Lawfulness, legitimacy, necessity, and data minimization	Classification and identifiability	PL2: Identifiable Personal Information	Information that can identify an individual, either alone or in combination, including direct identifiers and stable indirect identifiers	Disallowed by default	If strictly required for business purposes, store only in controlled systems with encryption, access control, and auditing; RAG should retain pointers rather than raw text	Avoid repetition or verbatim exposure in dialogue; when necessary, prompt for user confirmation and apply de-identification
Security safeguards and risk control	High-risk sensitive domains	PL3: Highly Sensitive Personal Information	Disclosure may materially harm personal safety, property, reputation, or health, or result in discriminatory treatment; typically requires a stronger legal basis	Not permitted	Only under compelling necessity and a clear legal basis may such data enter controlled storage; enforce strict minimization, field-level encryption, and robust auditing	Do not collect by default; if provided by the user, prioritize masking key fields and warning about the associated risks
Security safeguards and highest-level protection	Confidential information, credentials, directly actionable high-loss data	PL4: Confidential Information, Credentials	Once disclosed, can directly enable login, fund transfer, signature authorization, lateral movement, or expose undisclosed core business secrets	Absolutely prohibited	Prohibited from databases, memory, and logs; once detected, it must be redacted or blocked immediately; secrets must never enter model context	Must refuse collection, storage, or disclosure; advise credential rotation and escalation through secure incident-response procedures